

Supporting Information

Estimating potential-dependent physicochemical properties at metal–electrolyte interfaces using machine learning interatomic potentials

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System compositions and corresponding surface charge densities

Table S1: System compositions and surface charge densities ($\mu\text{C}/\text{cm}^2$) for Au(111), Cu(111), and Rh(111) surfaces.

System composition	Au(111)	Cu(111)	Rh(111)
94 water molecules and 2Cl^-	17.52	22.89	20.80
95 water molecules and 1Cl^-	8.76	11.45	10.40
96 water molecules	0.00	0.00	0.00
95 water molecules and 1Na^+	-8.76	-11.45	-10.40
94 water molecules and 2Na^+	-17.52	-22.89	-20.80
93 water molecules and 3Na^+	-26.28	-34.34	-31.20
92 water molecules and 4Na^+	-35.03	-45.78	-41.60

Learning rates for different MACE architectures

We perform a learning-rate sweep across the full training dataset to examine the convergence behavior of different MACE architectures using 840 test configuration. **Figure S1** shows how the energy and force mean absolute error (MAE) vary with total number of training configuration.

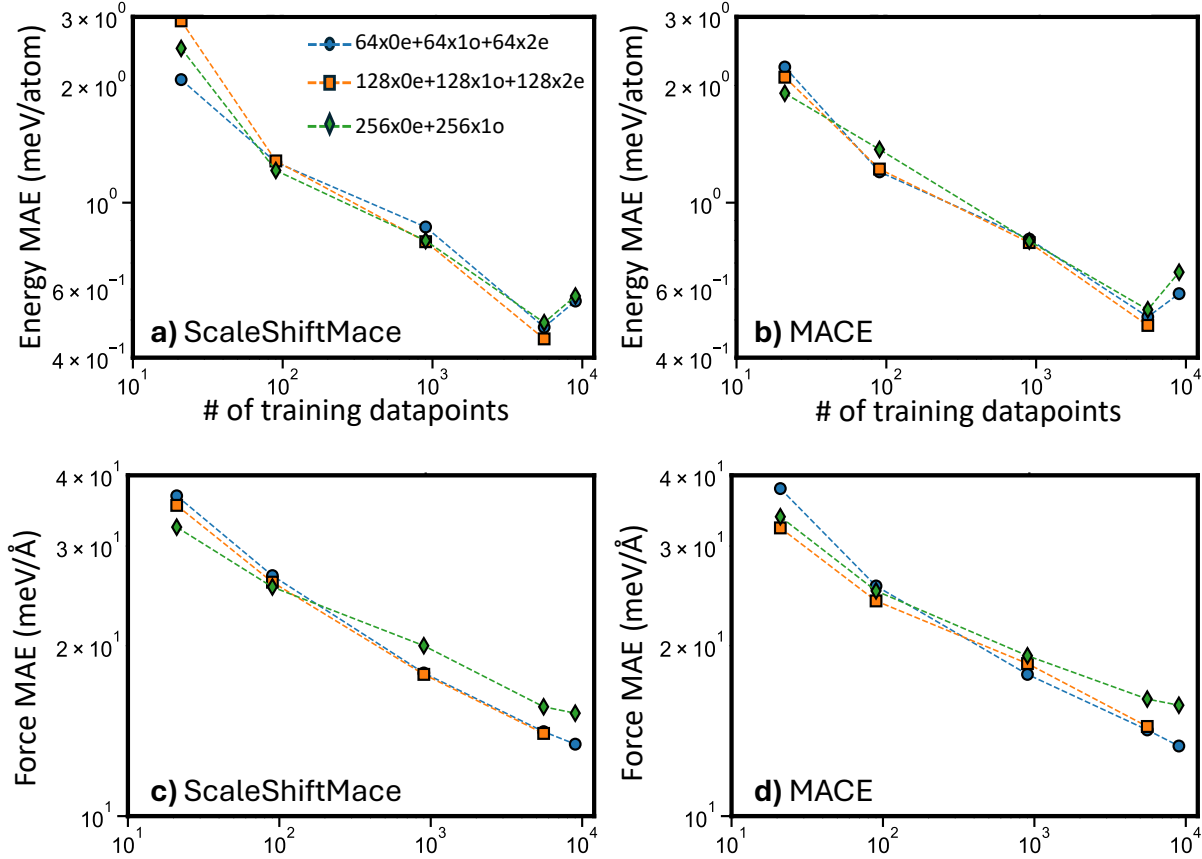


Figure S1: Learning rate sweep in log-log space to examine the convergence behavior of different MACE architectures. MAEs for (a-b) energy and (c-d) forces using the ScaleShiftMACE and MACE models for different hidden irreducible representations. The caption represents specific irreducible representations of the architecture.

Test set errors for different systems and elements

Table S2: Energy (meV/atom) and force (meV/Å) mean absolute errors on the test dataset for different systems given as mean $\pm$ std.

Dataset	Test set	Energy MAE [meV/atom]	Force MAE [meV/Å]
Au111 (17.52 $\mu\text{C}/\text{cm}^2$)	98	0.93 $\pm$ 0.02	18 $\pm$ 0.10
Au111 (8.76 $\mu\text{C}/\text{cm}^2$)	100	0.34 $\pm$ 0.11	12.72 $\pm$ 0.08
Au111 (0 $\mu\text{C}/\text{cm}^2$)	99	0.64 $\pm$ 0.12	12.75 $\pm$ 0.15
Au111 (−8.76 $\mu\text{C}/\text{cm}^2$)	99	0.45 $\pm$ 0.12	12.78 $\pm$ 0.13
Au111 (−17.52 $\mu\text{C}/\text{cm}^2$)	100	1.10 $\pm$ 0.14	12.33 $\pm$ 0.11
Au111 (−26.28 $\mu\text{C}/\text{cm}^2$)	100	0.98 $\pm$ 0.13	11.63 $\pm$ 0.15
Au111 (−35.03 $\mu\text{C}/\text{cm}^2$)	97	0.69 $\pm$ 0.06	18.23 $\pm$ 0.30
Cu111 (22.89 $\mu\text{C}/\text{cm}^2$)	99	0.54 $\pm$ 0.10	14.78 $\pm$ 0.16
Cu111 (11.45 $\mu\text{C}/\text{cm}^2$)	100	0.22 $\pm$ 0.01	14.10 $\pm$ 0.10
Cu111 (0 $\mu\text{C}/\text{cm}^2$)	99	0.77 $\pm$ 0.07	13.30 $\pm$ 0.19
Cu111 (−11.45 $\mu\text{C}/\text{cm}^2$)	100	0.24 $\pm$ 0.00	13.05 $\pm$ 0.11
Cu111 (−22.89 $\mu\text{C}/\text{cm}^2$)	100	0.56 $\pm$ 0.06	12.10 $\pm$ 0.07
Cu111 (−34.34 $\mu\text{C}/\text{cm}^2$)	100	0.55 $\pm$ 0.08	12.58 $\pm$ 0.19
Cu111 (−45.78 $\mu\text{C}/\text{cm}^2$)	100	0.36 $\pm$ 0.02	11.55 $\pm$ 0.22
Rh111 (20.80 $\mu\text{C}/\text{cm}^2$)	97	0.51 $\pm$ 0.11	16.40 $\pm$ 0.16
Rh111 (10.40 $\mu\text{C}/\text{cm}^2$)	100	0.23 $\pm$ 0.06	15.43 $\pm$ 0.28
Rh111 (0 $\mu\text{C}/\text{cm}^2$)	100	0.59 $\pm$ 0.13	15.13 $\pm$ 0.19
Rh111 (−10.40 $\mu\text{C}/\text{cm}^2$)	98	0.17 $\pm$ 0.01	14.18 $\pm$ 0.08
Rh111 (−20.80 $\mu\text{C}/\text{cm}^2$)	83	0.24 $\pm$ 0.06	14.80 $\pm$ 0.12
Rh111 (−31.20 $\mu\text{C}/\text{cm}^2$)	87	0.45 $\pm$ 0.09	14.20 $\pm$ 0.10
Rh111 (−41.60 $\mu\text{C}/\text{cm}^2$)	93	0.20 $\pm$ 0.02	13.78 $\pm$ 0.29
Au111	697	0.73 $\pm$ 0.04	14.02 $\pm$ 0.03
Cu111	698	0.46 $\pm$ 0.01	13.05 $\pm$ 0.09
Rh111	658	0.34 $\pm$ 0.03	14.88 $\pm$ 0.18
All	2053	0.52 $\pm$ 0.02	14 $\pm$ 0.05

Table S3: Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) for per element forces on test set in units of meV/Å.

Element	RMSE (meV/Å)	MAE (meV/Å)
Au	10.745	7.925
Cl	88.134	42.973
Cu	8.713	6.471
H	19.125	12.637
Na	43.872	28.461
O	33.235	21.355
Rh	17.339	13.158

Benchmarking MACE-MD predicted density of water and ions for different systems against AIMD

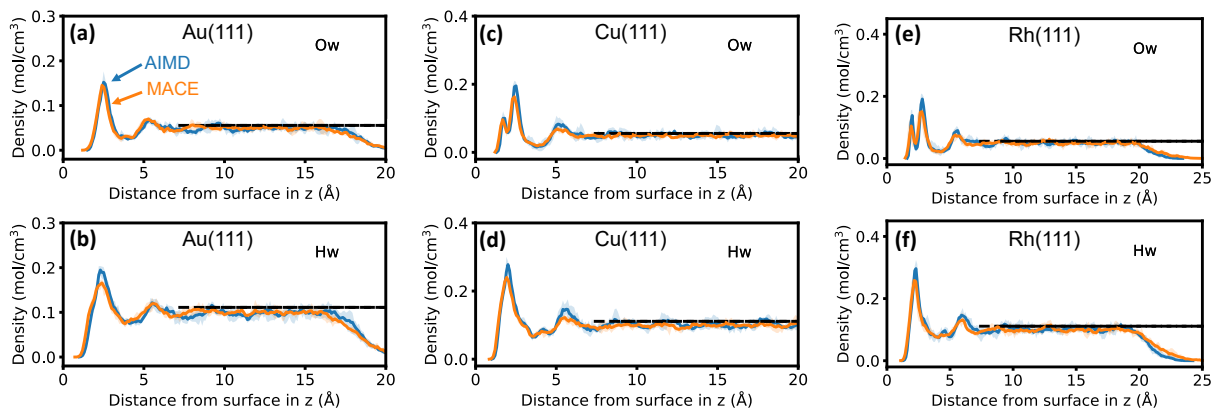


Figure S2: Atomic density profiles of oxygen and hydrogen of the water molecules on (a-b) Au(111), (c-d) Cu(111), and (e-f) Rh(111) from AIMD and MACE-MD at potential of zero charge. Each density profile is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean. The black line denotes the bulk water density.

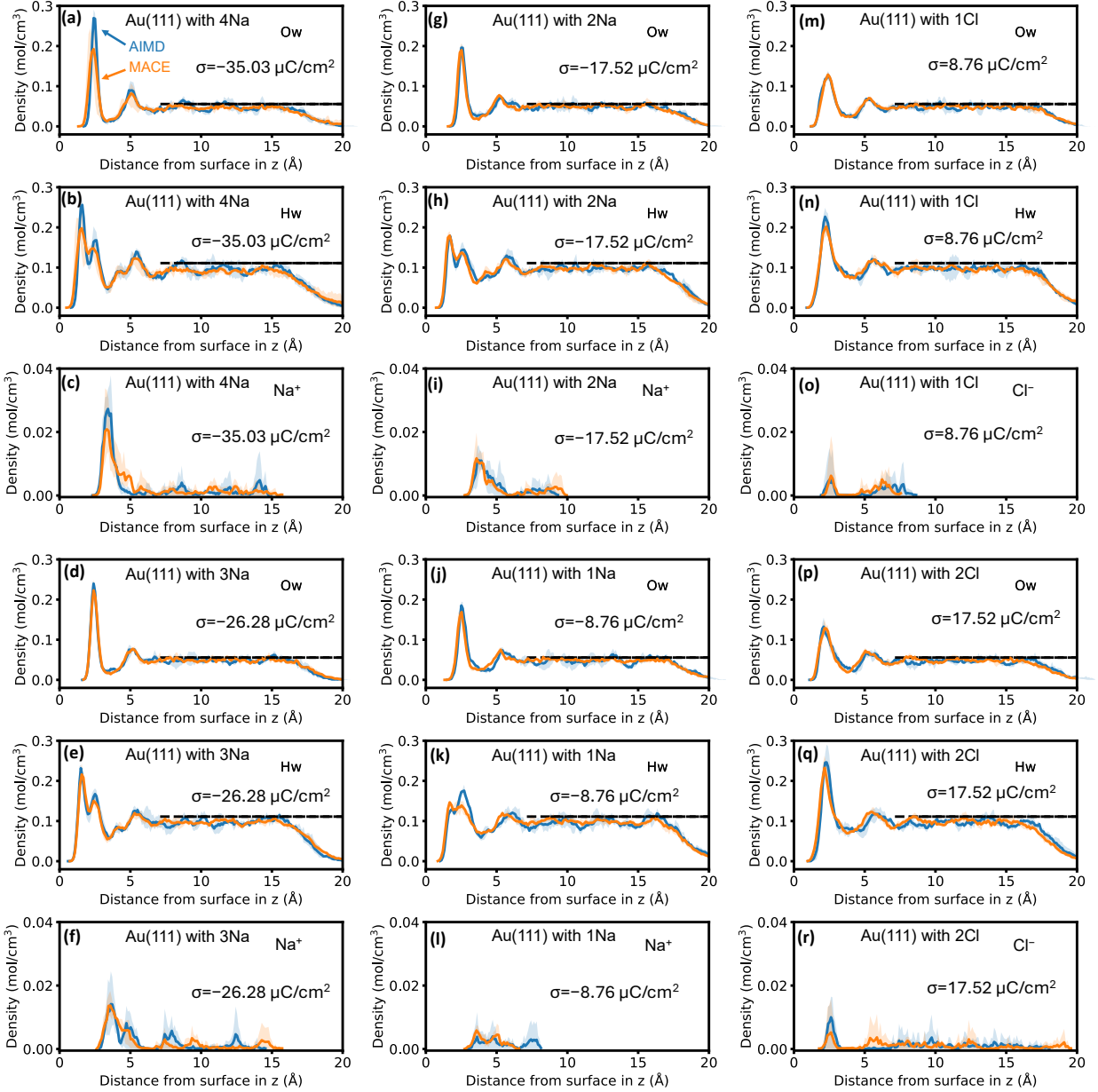


Figure S3: Atomic density profiles of oxygen and hydrogen of the water molecules and that of the ions on Au(111) from AIMD and MACE-MD for surface charge density **(a-c)** $-35.03 \mu\text{C}/\text{cm}^2$, **(d-f)** $-26.28 \mu\text{C}/\text{cm}^2$, **(g-i)** $-17.52 \mu\text{C}/\text{cm}^2$, **(j-l)** $-8.76 \mu\text{C}/\text{cm}^2$, **(m-o)** $8.76 \mu\text{C}/\text{cm}^2$, **(p-r)** $17.52 \mu\text{C}/\text{cm}^2$. Each density profile is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean. The black line denotes the bulk water density.

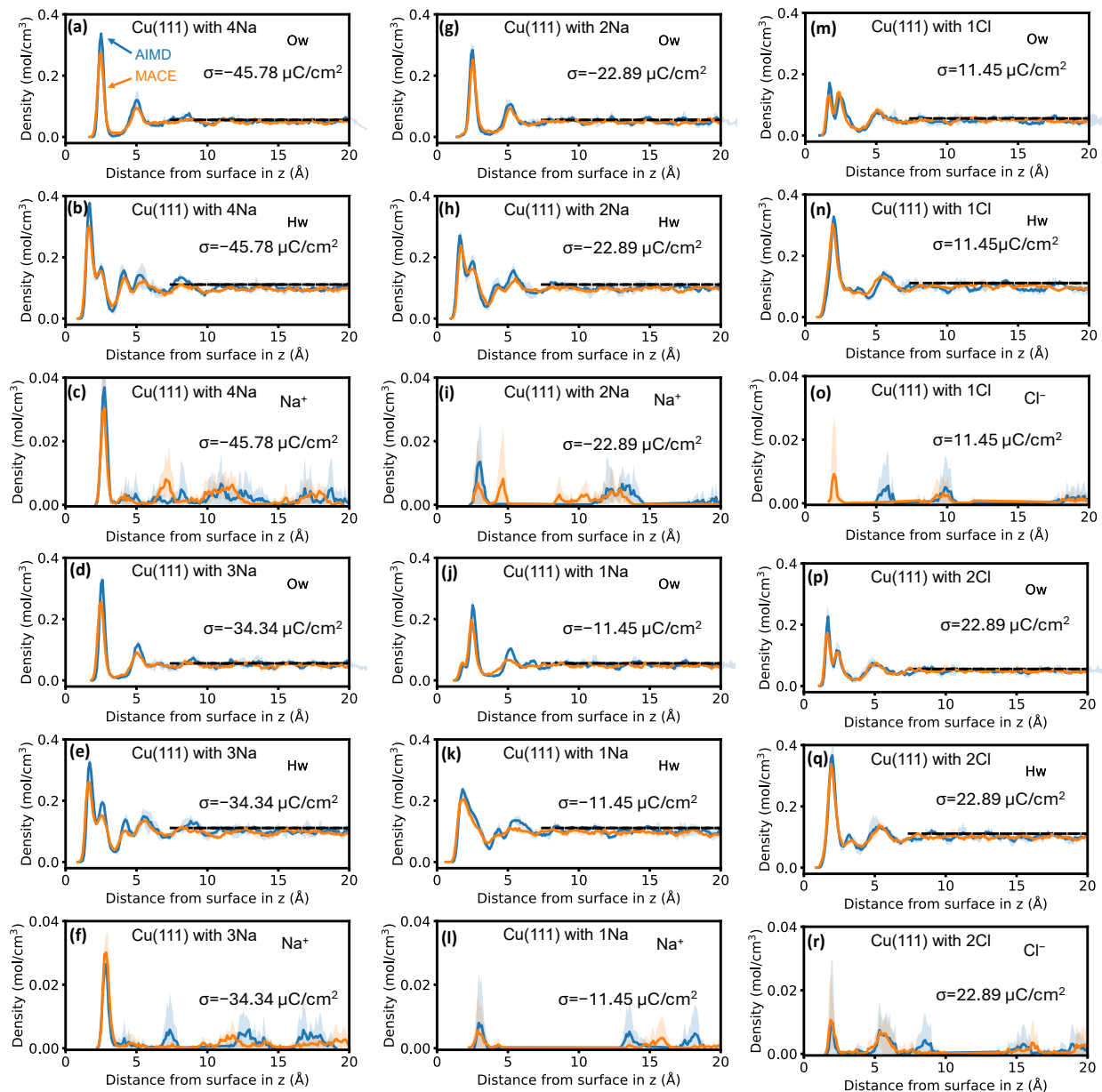


Figure S4: Atomic density profiles of oxygen and hydrogen of the water molecules and that of the ions on Cu(111) from AIMD and MACE-MD for surface charge density (a-c) $-45.78 \mu\text{C}/\text{cm}^2$, (d-f) $-34.34 \mu\text{C}/\text{cm}^2$, (g-i) $-22.89 \mu\text{C}/\text{cm}^2$, (j-l) $-11.45 \mu\text{C}/\text{cm}^2$, (m-o) $11.45 \mu\text{C}/\text{cm}^2$, (p-r) $22.89 \mu\text{C}/\text{cm}^2$. Each density profile is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean. The black line denotes the bulk water density.

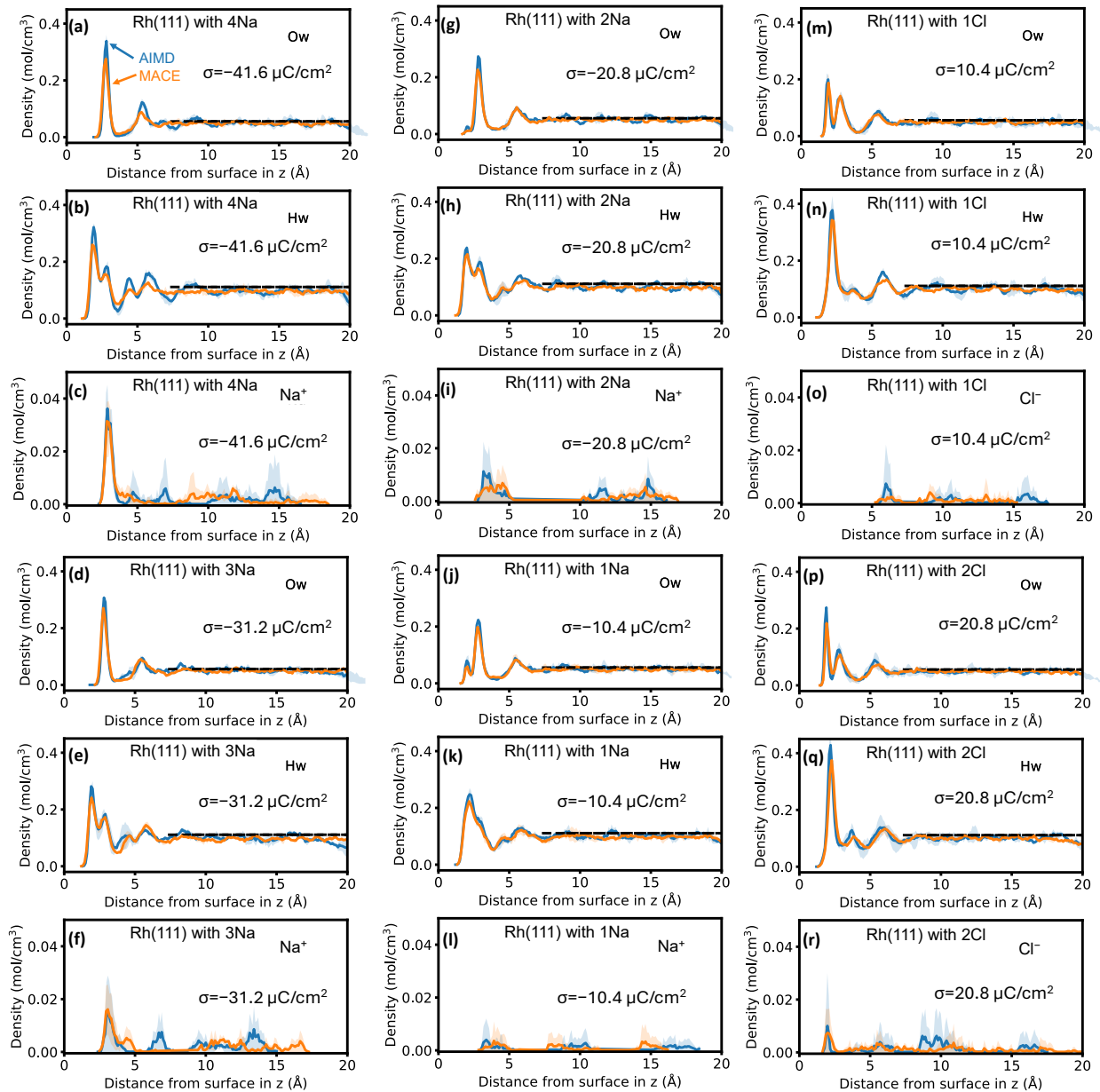


Figure S5: Atomic density profiles of oxygen and hydrogen of the water molecules and that of the ions on Rh(111) from AIMD and MACE-MD for surface charge density (a-c) $-41.6 \mu\text{C}/\text{cm}^2$, (d-f) $-31.2 \mu\text{C}/\text{cm}^2$, (g-i) $-20.8 \mu\text{C}/\text{cm}^2$, (j-l) $-10.4 \mu\text{C}/\text{cm}^2$, (m-o) $10.4 \mu\text{C}/\text{cm}^2$, (p-r) $20.8 \mu\text{C}/\text{cm}^2$. Each density profile is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean. The black line denotes the bulk water density.

Benchmarking MACE-MD predicted orientational distribution of water against AIMD

Table S4: Distances (in Å) used to identify water molecules in the chemisorption region and in the first and second water layers for calculating their orientational distribution.

Metal	Avg. surf. dist.	$r_{\text{H}_2\text{O,Me}}$	Chemisorbed layer	First layer	Second layer
Au(111)	7.4	2.60	10	10.78	14.1
Cu(111)	6.28	2.55	8.83	9.73	12.7
Rh(111)	6.54	2.68	9.22	10.24	13.54

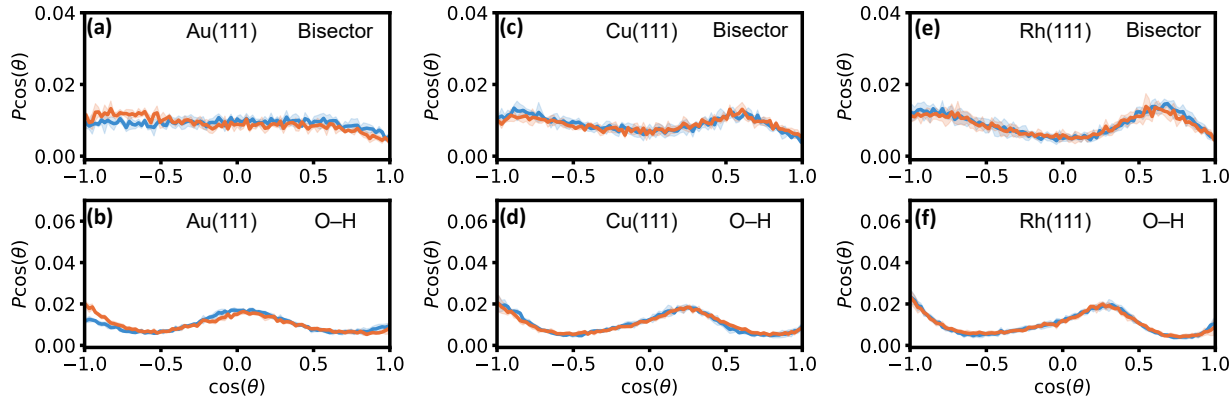


Figure S6: Orientational distribution of the first water layer measured as the probability distribution of the cosines of the angle between the bisector of the water and the surface normal and between the O-H bonds of water and the surface normal for **(a-b)** Au, **(c-d)** Cu, and **(e-f)** Rh (111) at the potential of zero charge using AIMD and MACE-MD. Each line is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean.

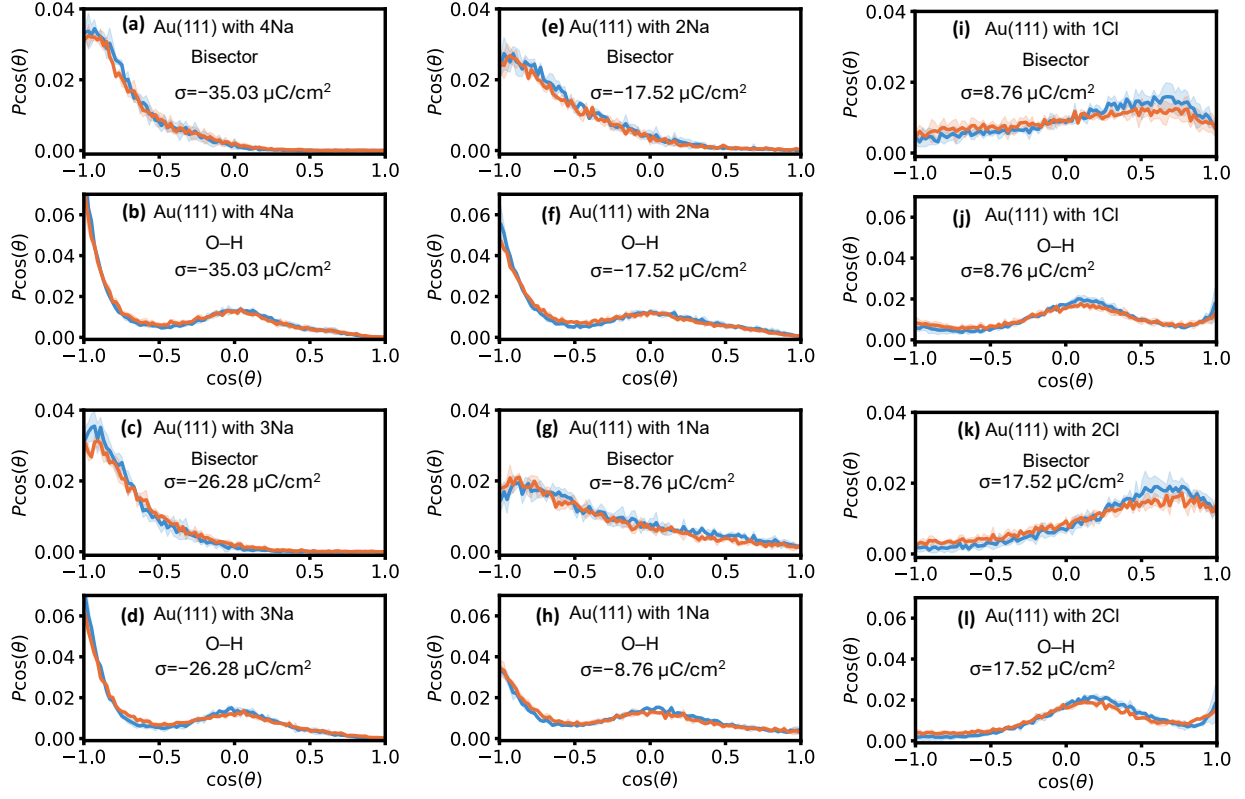


Figure S7: Orientational distribution of the first water layer measured as the probability distribution of the cosines of the angle between the bisector of the water and the surface normal and between the O-H bonds of water and the surface normal using AIMD and MACE-MD for Au(111) system with surface charge density of (a-b) $-35.03 \mu\text{C}/\text{cm}^2$, (c-d) $-26.28 \mu\text{C}/\text{cm}^2$, (e-f) $-17.52 \mu\text{C}/\text{cm}^2$, (g-h) $-8.76 \mu\text{C}/\text{cm}^2$, (i-j) $8.76 \mu\text{C}/\text{cm}^2$, (k-l) $17.52 \mu\text{C}/\text{cm}^2$. Each line is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean.

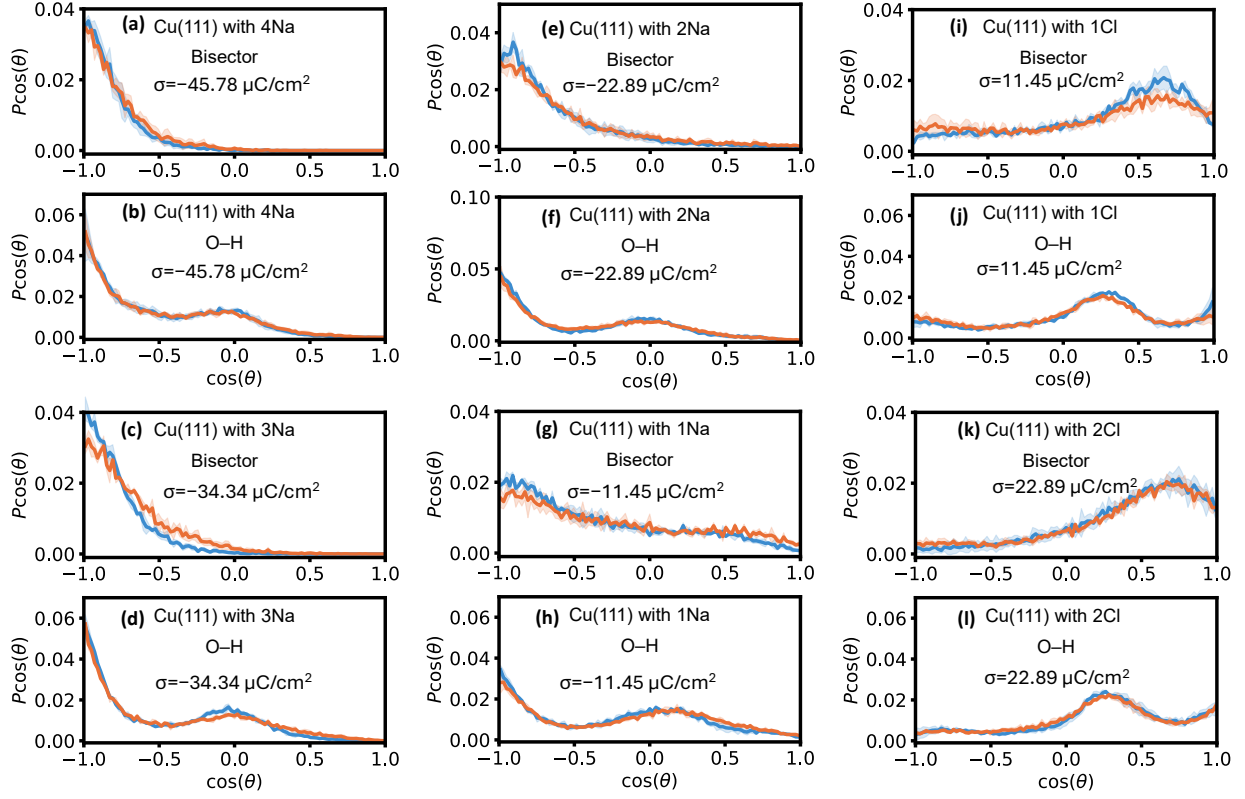


Figure S8: Orientational distribution of the first water layer measured as the probability distribution of the cosines of the angle between the bisector of the water and the surface normal and between the O–H bonds of water and the surface normal using AIMD and MACE-MD for Cu(111) system with surface charge density of **(a-b)** $-35.03 \mu\text{C}/\text{cm}^2$, **(c-d)** $-26.28 \mu\text{C}/\text{cm}^2$, **(e-f)** $-17.52 \mu\text{C}/\text{cm}^2$, **(g-h)** $-8.76 \mu\text{C}/\text{cm}^2$, **(i-j)** $8.76 \mu\text{C}/\text{cm}^2$, **(k-l)** $17.52 \mu\text{C}/\text{cm}^2$. Each line is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean.

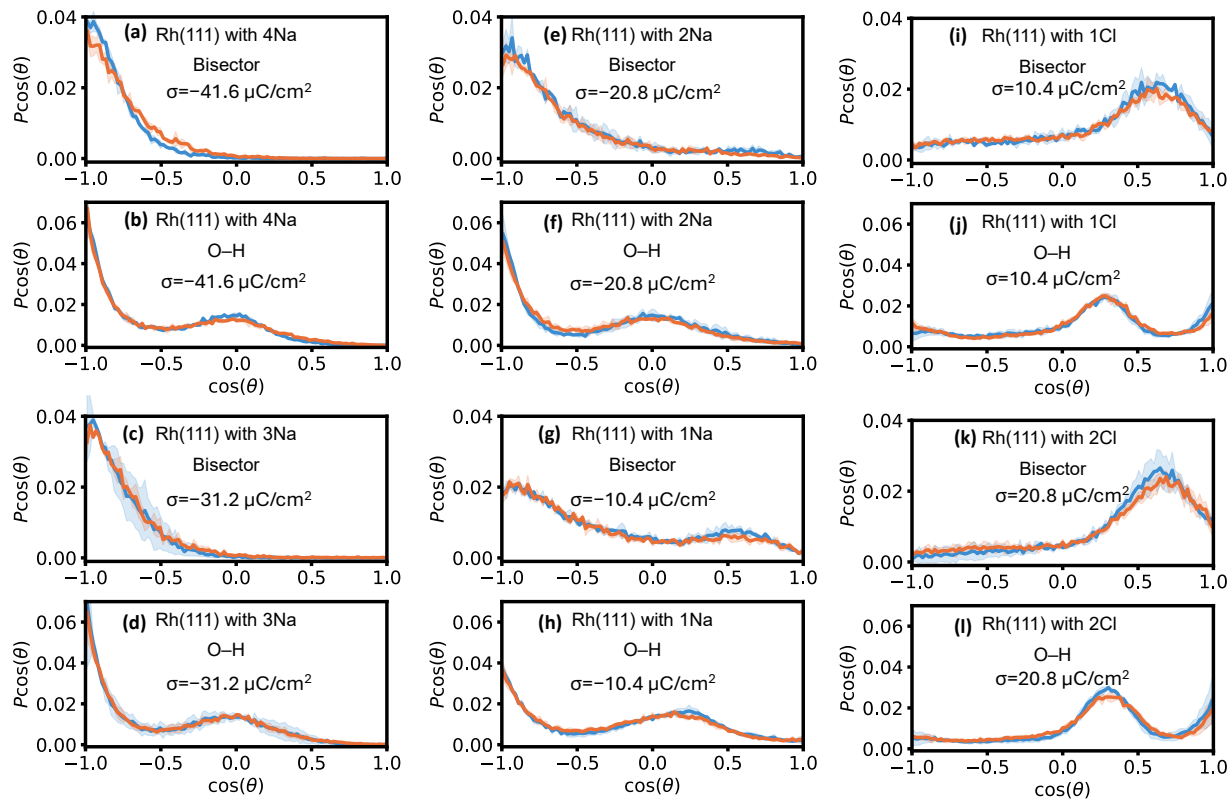


Figure S9: Orientational distribution of the first water layer measured as the probability distribution of the cosines of the angle between the bisector of the water and the surface normal and between the O-H bonds of water and the surface normal using AIMD and MACE-MD for Rh(111) system with surface charge density of (a-b) $-35.03 \mu\text{C}/\text{cm}^2$, (c-d) $-26.28 \mu\text{C}/\text{cm}^2$, (e-f) $-17.52 \mu\text{C}/\text{cm}^2$, (g-h) $-8.76 \mu\text{C}/\text{cm}^2$, (i-j) $8.76 \mu\text{C}/\text{cm}^2$, (k-l) $17.52 \mu\text{C}/\text{cm}^2$. Each line is an average of three separate 10 ps runs and the shaded region indicates one standard deviation away from mean.

Cosine similarity score and Pearson correlation coefficient values for comparing MACE-MD against AIMD

Table S5: Average cosine similarity score and Pearson correlation coefficient values between the water coverage for different systems.

System	Cosine similarity	Pearson correlation
Au111	1.000	0.999
Cu111	0.997	0.997
Rh111	0.998	0.996

Table S6: Cosine similarity score and Pearson correlation coefficient for angle bisector and O-H bond of the interfacial water for different electrolyte-electrode systems.

System	Au(111)				Cu(111)				Rh(111)			
	bisector		OH		bisector		OH		bisector		OH	
	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r
2Cl	0.985	0.978	0.988	0.973	0.994	0.984	0.995	0.983	0.993	0.990	0.991	0.981
1Cl	0.982	0.930	0.990	0.976	0.981	0.948	0.990	0.961	0.995	0.985	0.992	0.967
water	0.983	0.460	0.985	0.864	0.994	0.873	0.996	0.980	0.995	0.955	0.998	0.989
1Na	0.990	0.975	0.992	0.970	0.983	0.970	0.991	0.972	0.995	0.983	0.996	0.985
2Na	0.996	0.992	0.993	0.987	0.994	0.991	0.994	0.990	0.996	0.993	0.993	0.985
3Na	0.993	0.992	0.993	0.992	0.975	0.974	0.994	0.990	0.996	0.995	0.998	0.997
4Na	0.997	0.995	0.995	0.993	0.994	0.994	0.998	0.996	0.989	0.989	0.996	0.994
Average	0.989	0.903	0.991	0.965	0.988	0.962	0.994	0.982	0.994	0.984	0.995	0.985

Table S7: Cosine similarity score and Pearson correlation coefficient for the density peaks of interfacial water (oxygen and hydrogen) for different electrolyte-electrode systems.

System	Au(111)				Cu(111)				Rh(111)			
	Oxygen		Hydrogen		Oxygen		Hydrogen		Oxygen		Hydrogen	
	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r	$\cos(\alpha)$	r
2Cl	0.976	0.966	0.974	0.851	1.000	1.000	0.999	0.997	1.000	1.000	0.934	0.547
1Cl	0.992	0.993	0.999	0.999	0.985	0.918	0.999	0.999	1.000	1.000	0.995	0.979
water	0.997	0.991	1.000	1.000	1.000	1.000	0.999	0.999	1.000	1.000	0.999	0.997
1Na	0.982	0.973	0.990	1.000	0.999	1.000	1.000	0.998	1.000	1.000	1.000	0.998
2Na	1.000	1.000	1.000	1.000	0.989	0.993	1.000	1.000	0.998	0.992	1.000	1.000
3Na	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.989	0.995
4Na	0.983	0.982	1.000	1.000	0.994	0.990	1.000	1.000	1.000	1.000	1.000	1.000
Average	0.990	0.986	0.995	0.978	0.995	0.986	1.000	0.999	0.999	0.998	0.988	0.931

AIMD and MACE-MD predicted work function, dipoles, and net water orientation on Au(111)

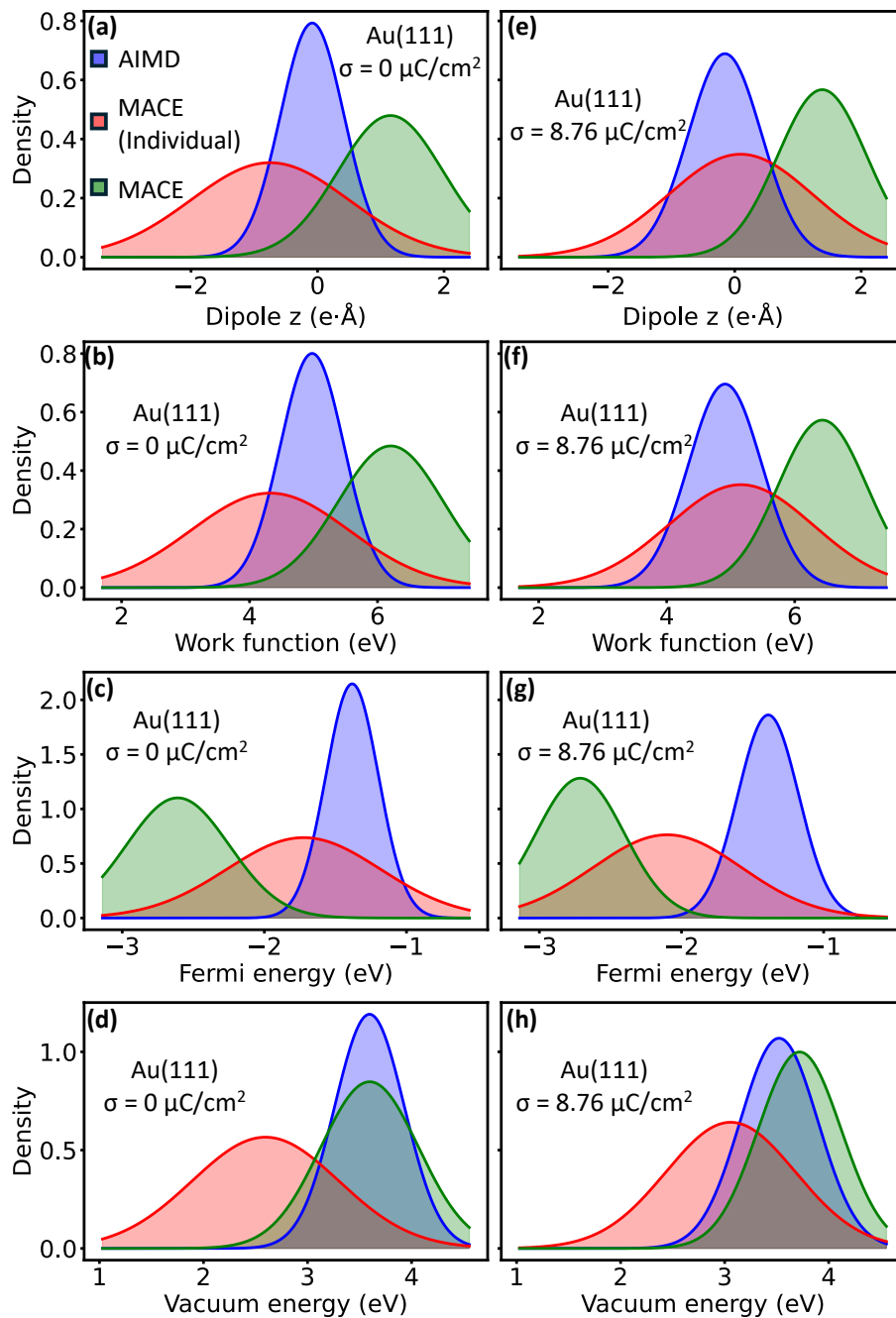


Figure S10: Normalized gaussian distributions of dipole moment (z -direction), work function, fermi energy, and vacuum energy for Au(111) obtained from AIMD and MACE-MD snapshot single-point calculations at (a–d) $\sigma = 0 \mu\text{C}/\text{cm}^2$ and (e–h) $\sigma = 8.76 \mu\text{C}/\text{cm}^2$. The two MACE-MD distributions are from MLIPs trained on individual system dataset and entire dataset.

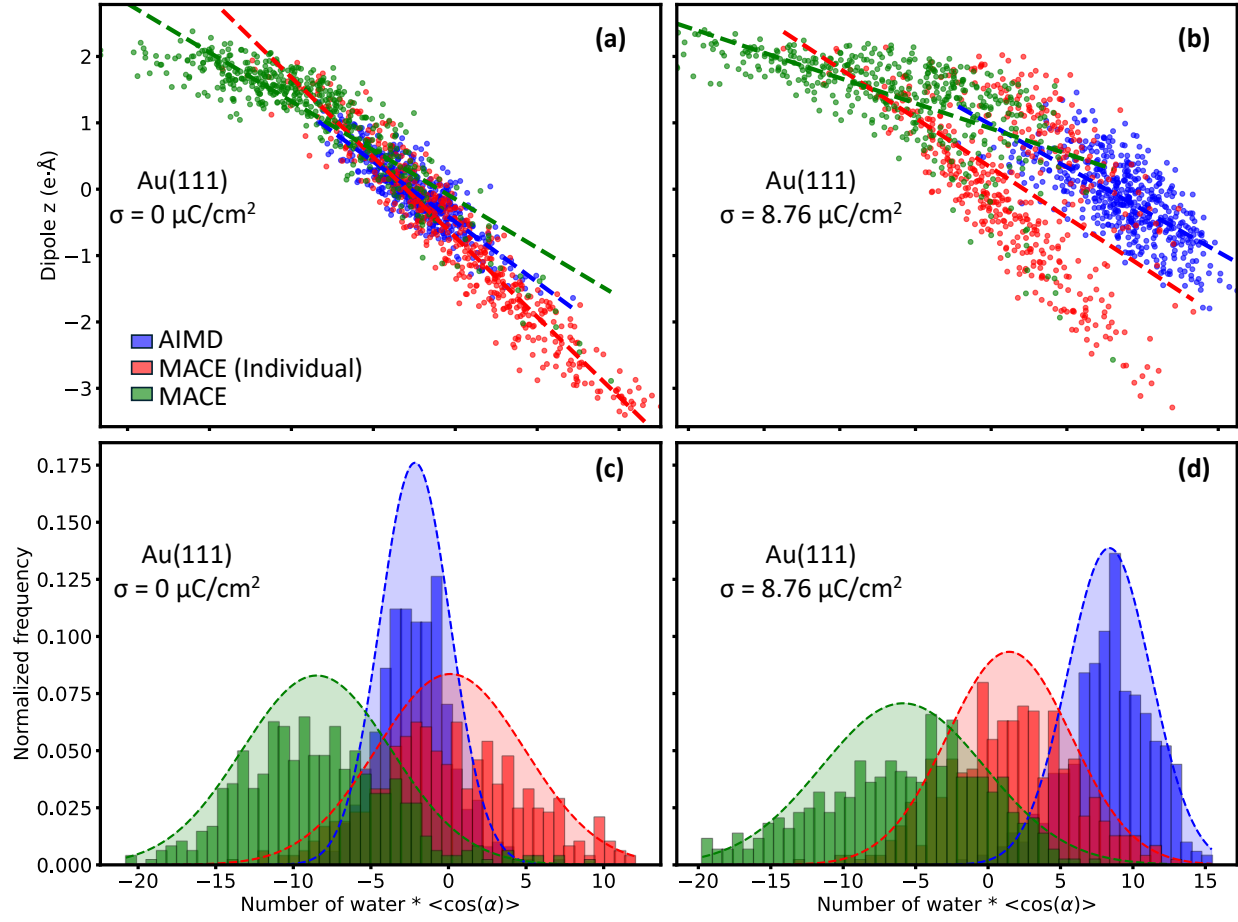


Figure S11: Comparison of dipole moment (z -direction) and average water orientation on Au(111) obtained from AIMD and MACE-MD snapshot single-point calculations at (a) $\sigma = 0 \mu\text{C}/\text{cm}^2$ and (b) $\sigma = 8.76 \mu\text{C}/\text{cm}^2$. Normalized distributions of average water orientation for Au(111) obtained from AIMD and MACE-MD snapshot single-point calculations at (c) $\sigma = 0 \mu\text{C}/\text{cm}^2$ and (d) $\sigma = 8.76 \mu\text{C}/\text{cm}^2$. The two MACE-MD distributions are from MLIPs trained on individual system dataset and entire dataset.

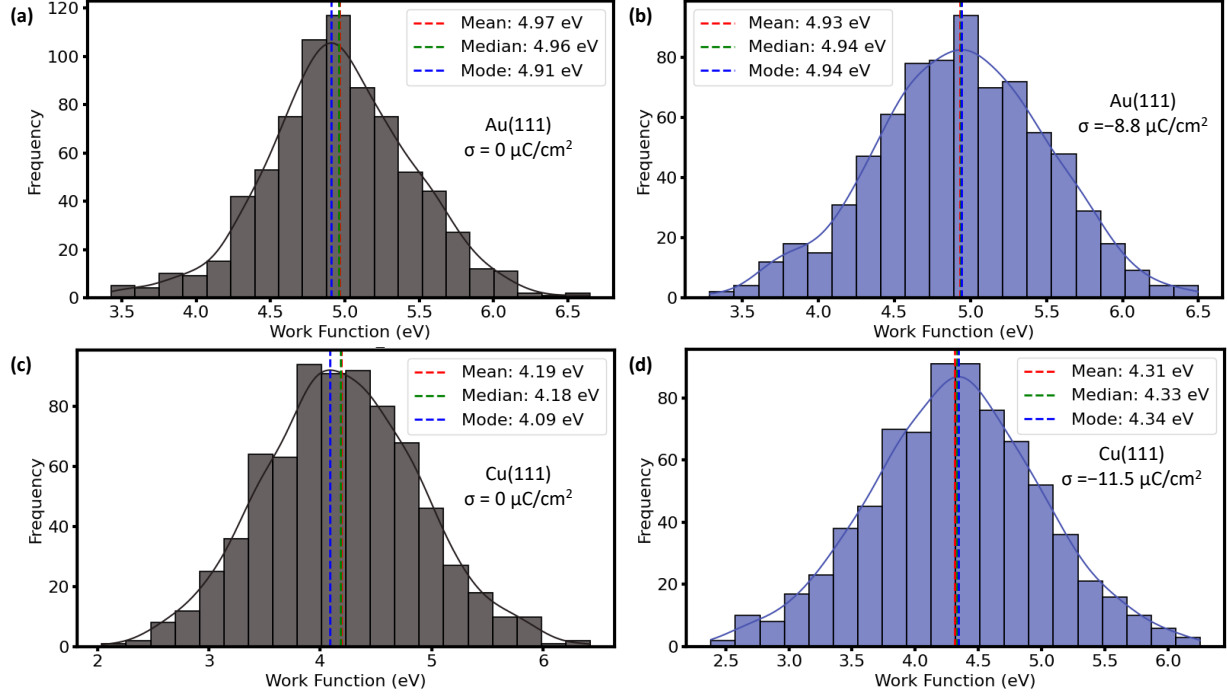


Figure S12: Normalized Gaussian distributions of the work function for Au(111) at (a) $\sigma = 0 \mu\text{C}/\text{cm}^2$ and (b) $\sigma = -8.8 \mu\text{C}/\text{cm}^2$, and for Cu(111) at (c) $\sigma = 0 \mu\text{C}/\text{cm}^2$ and (d) $\sigma = -11.5 \mu\text{C}/\text{cm}^2$. Work functions were obtained from single-point calculations on AIMD snapshots sampled every 40 fs from three independent 10 ps simulations.

Convergence test for coverage of oxygen of water

Table S8: Calculated water coverages on the Pt(111)-water interface from previous reports. All studies are based on AIMD simulations or neural network potentials (NNP)-MD trained on AIMD data.(Adapted from AIP Publishing¹)

Study	Functional	Surface Size	θ_w (ML)	Ref.
Liu <i>et al.</i>	RPBE-D3	4×4	0.135 ± 0.004	¹
Kelly <i>et al.</i>	RPBE-D3	4×3	0.15	²
Kristoffersen <i>et al.</i>	RPBE-D3	4×3	0.16	³
Le <i>et al.</i>	PBE-D3	4×4	0.16	⁴
Le <i>et al.</i>	PBE-D3	6×6	0.18	⁵
Mikkelsen <i>et al.</i>	RPBE-D3 (NNP)	12×16	0.15	⁶

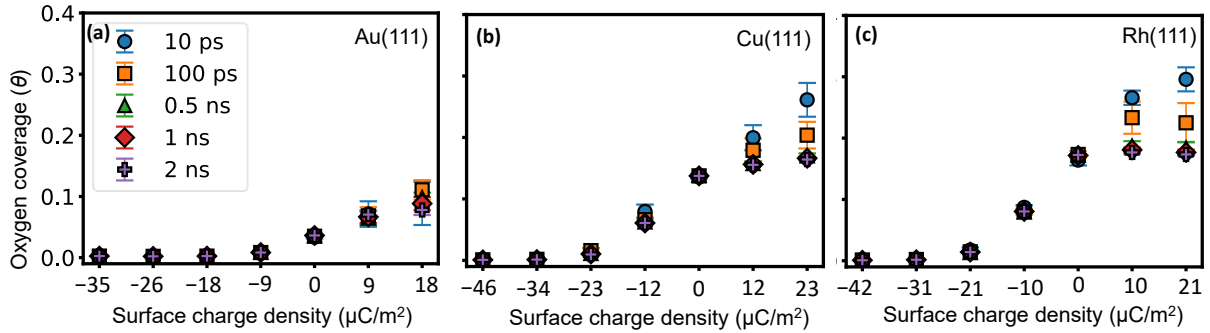


Figure S13: Surface charge density versus oxygen coverage on (a) Au(111), (b) Cu(111), and (c) Rh(111) for different simulation timescales. Each scatter point represents an average oxygen coverage. Each result is calculated from three independent 2 ns MACE-MD simulations sampled every 20 fs.

Coverage of water oxygen and hydrogen and ion coverage on Au(111) and Cu(111) at potential of zero charge

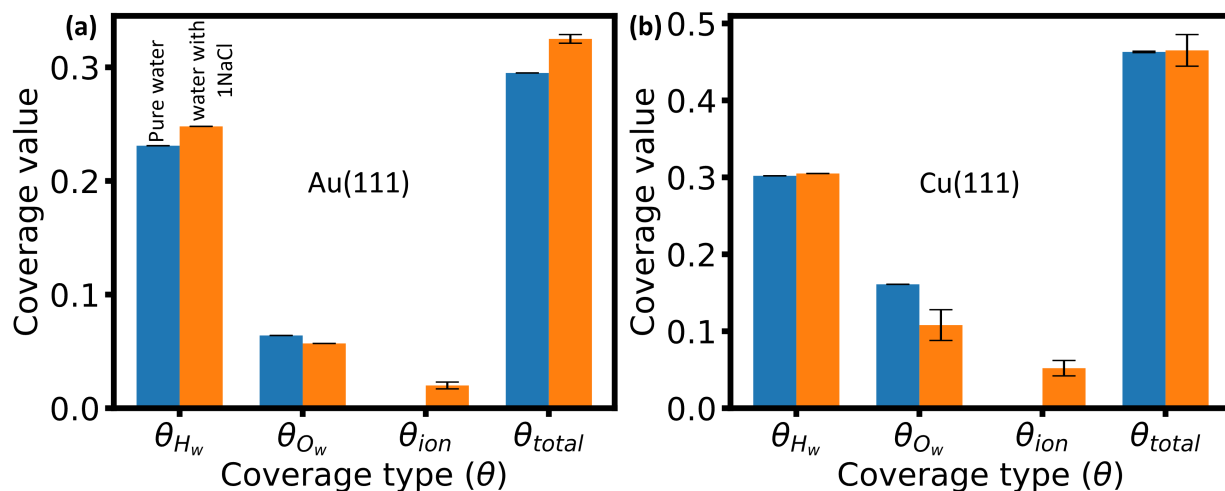


Figure S14: Surface coverage of water oxygen and hydrogen atoms, ion coverage, and total coverage on Au(111) and Cu(111). Coverage of water oxygen (θ_{O_w}) and hydrogen (θ_{H_w}), ion coverage (θ_{ions}), and total coverage (θ_{total}) are shown for (a) Au(111) and (b) Cu(111) at the potential of zero charge for system containing pure water or water with one NaCl. Coverage values are computed using the method described in section 2.4.4. Error bars, which are small, denotes one standard deviation of coverage from three independent MACE-MD simulations sampled every 20 fs. Potentials are calculated by averaging results from three independent 10 ps AIMD simulations sampled at 100 fs intervals.

Radial distribution function to study the water distribution near ions

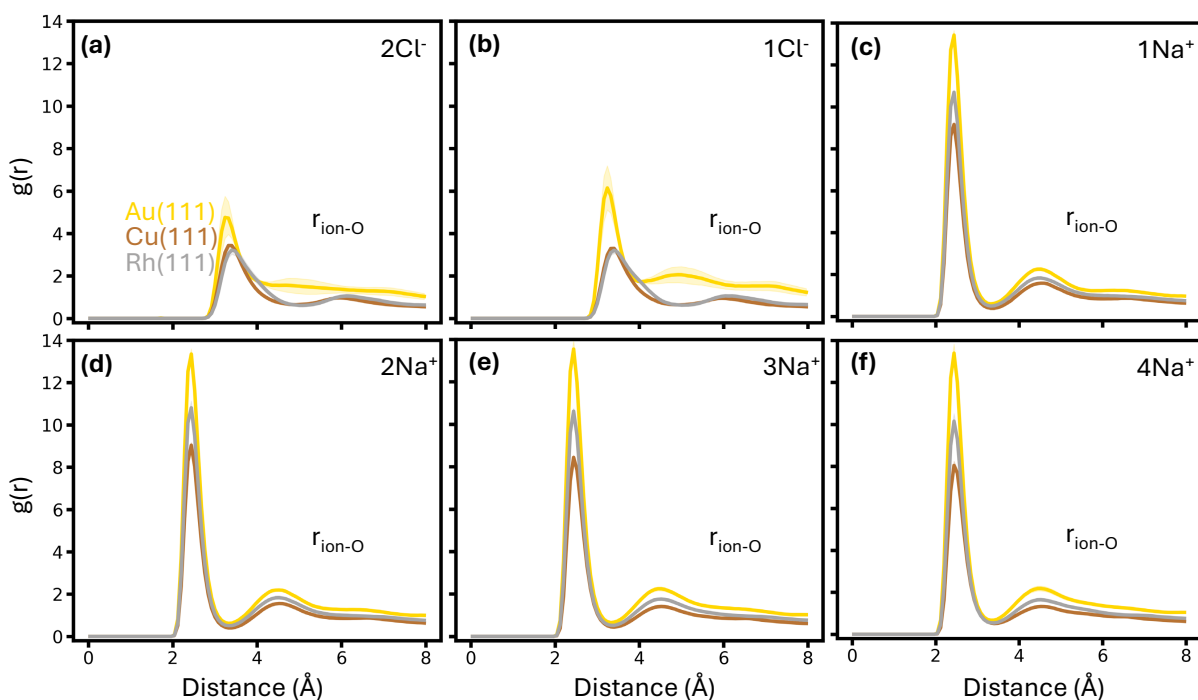


Figure S15: Radial distribution functions ($g(r)$) of H_w around Na^+ and Cl^- on different metals for systems with (a) $2Cl^-$, (b) $1Cl^-$, (c) $1Na^+$, (d) $2Na^+$, (e) $3Na^+$, and (f) $4Na^+$ using 2 ns MACE-MD. Each solid line is an average of three separate 2 ns runs, and the shaded region indicates one standard deviation away from the mean.

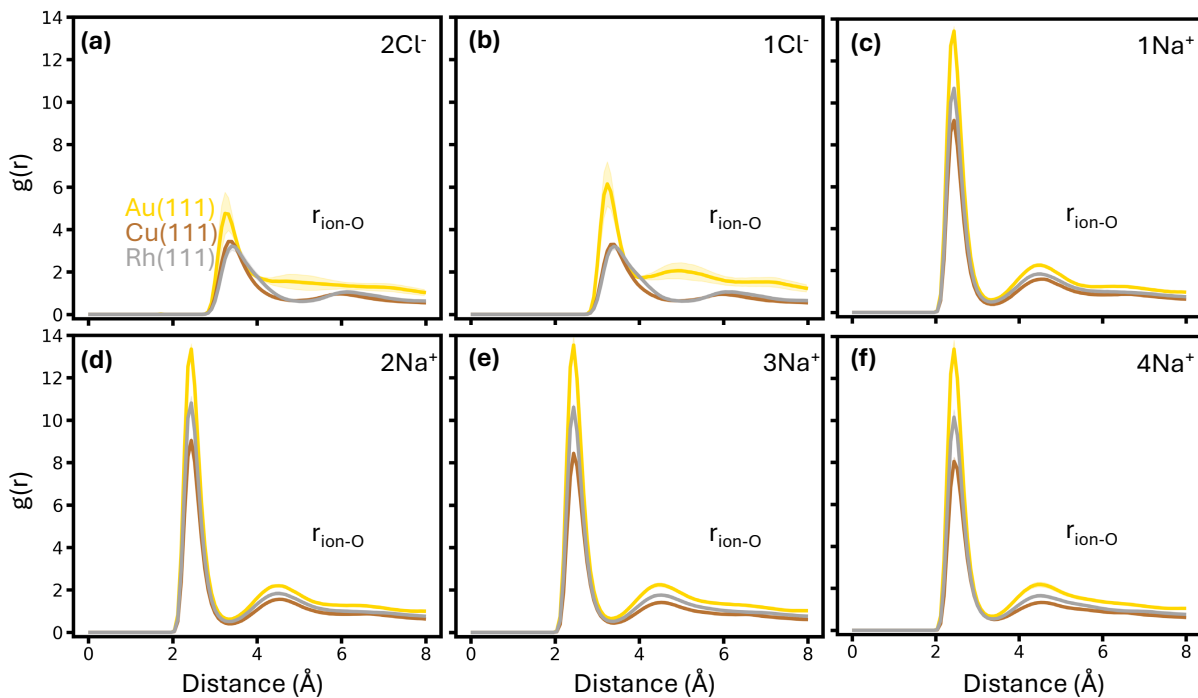


Figure S16: Radial distribution functions ($g(r)$) of O_w around Na^+ and Cl^- on different metals for systems with (a) $2Cl^-$, (b) $1Cl^-$, (c) $1Na^+$, (d) $2Na^+$, (e) $3Na^+$, and (f) $4Na^+$ using 2 ns MACE-MD. Each solid line is an average of three separate 2 ns runs, and the shaded region indicates one standard deviation away from the mean.

Variation of ion distance and oxygen coverage with surface charge density on different metals

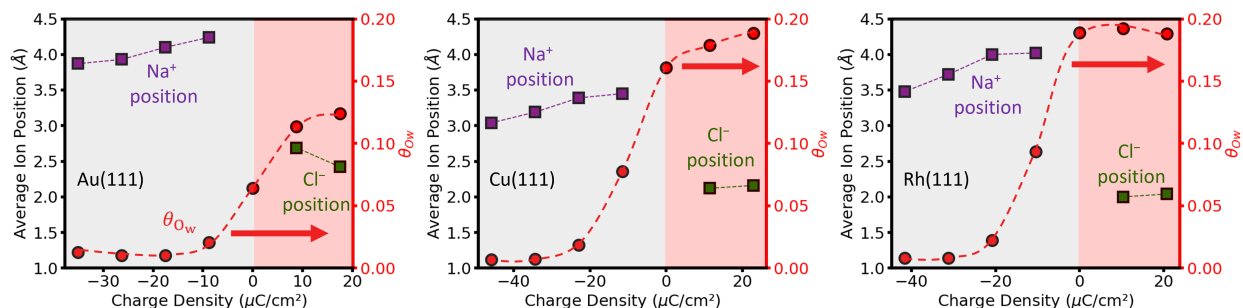


Figure S17: Surface charge density versus average ion position near surface and water oxygen coverage on (a) Au(111), (b) Cu(111), and (c) Rh(111). Each scatter point represents an average ion position or oxygen coverage, and the dashed lines connecting the points are drawn to guide the eye. Each result is calculated from three independent 2 ns MACE-MD simulations sampled every 20 fs.

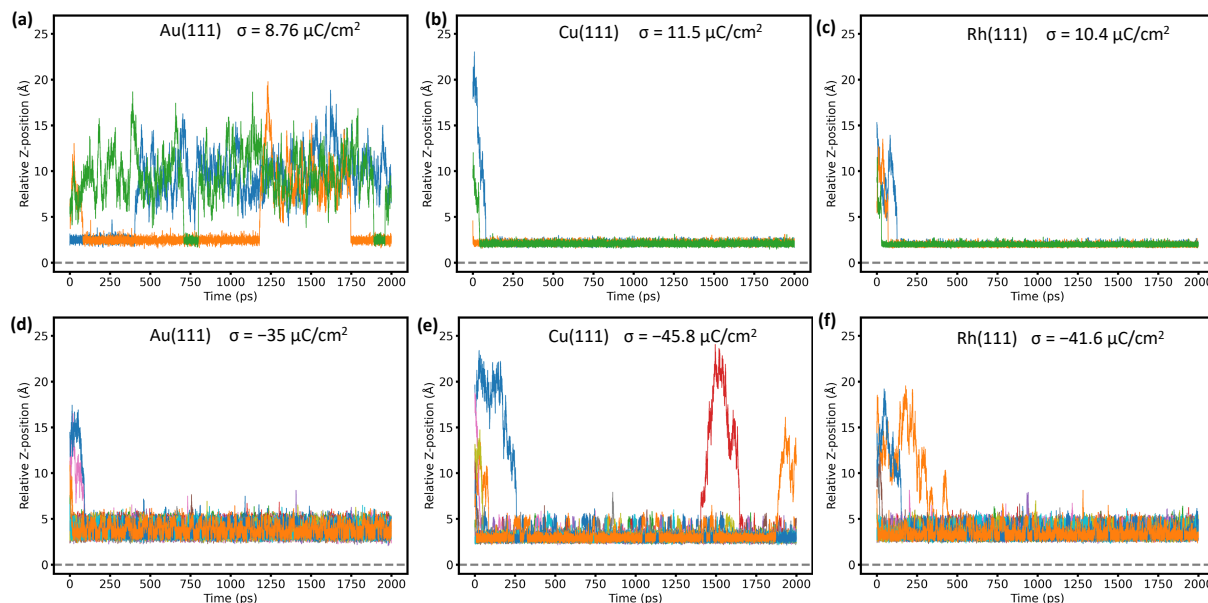


Figure S18: Dynamic trajectories of interfacial ions on charged metal surfaces over 2 ns MACE-MD. Cl⁻ position on (a) Au(111), (b) Cu(111) and (c) Rh(111), and Na⁺ position on (d) Au(111), (e) Cu(111), and (f) Rh(111). Each uniquely colored line represents the continuous trajectory of an individual ion over the 2 ns simulation.

S12 Complete atomic density profiles of water and ions on different metals

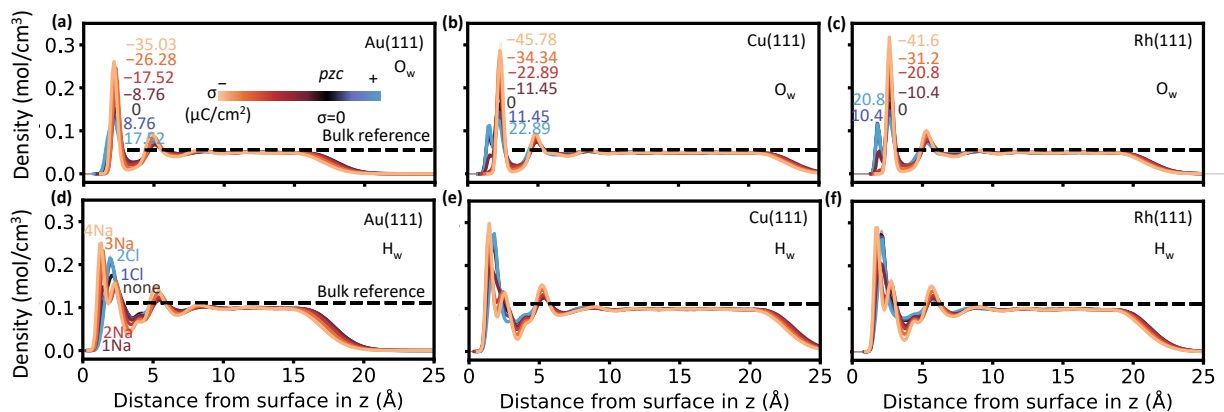


Figure S19: Atomic density profiles of **(a-c)** oxygen and **(d-f)** hydrogen of water molecules and on Au, Cu, and Rh (111) surfaces from MACE-MD. Each solid line represents an atomic density for a specific ion concentration (surface charge density), and the heatmap given in **(a)** shows the variation in surface charge density with ion type and number of ions. Each solid density profile is an average of three independent 2 ns MACE-MD simulations, and the shaded region indicates one standard deviation away from the mean. For an atomic density, the mean density is represented by a solid line, while the statistical variance across the simulations is shown by a shaded region spanning the minimum and maximum values. The coordinate system is referenced such that $z = 0$ corresponds to the position of the uppermost metallic layer, and horizontal lines indicating the expected bulk densities of solvent oxygen and hydrogen are included for comparison. The dashed black line represents the bulk oxygen or hydrogen density for water.

Orientational distribution of water in the chemisorbed layer, layer one, and layer two using MACE-MD

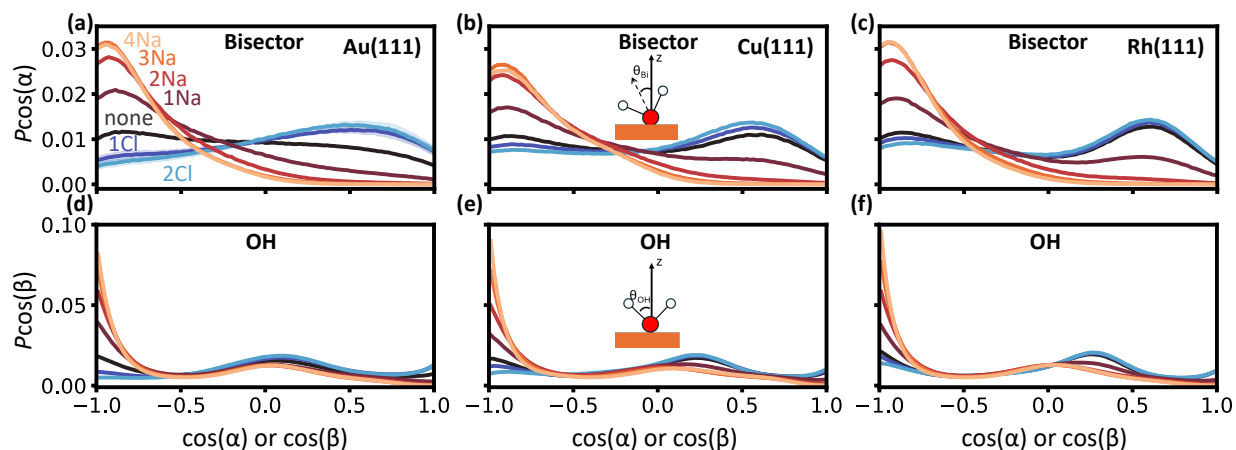


Figure S20: Orientational distribution of the first water layer measured as the probability distribution of the cosines of the angle between **(a-c)** the bisector of the water and the surface normal and **(d-f)** between the O-H bonds of water and the surface normal for Au, Cu, and Rh (111) facets using MACE-MD. Each solid line is an average of three separate 2 ns runs, and the shaded region indicates one standard deviation away from the mean.

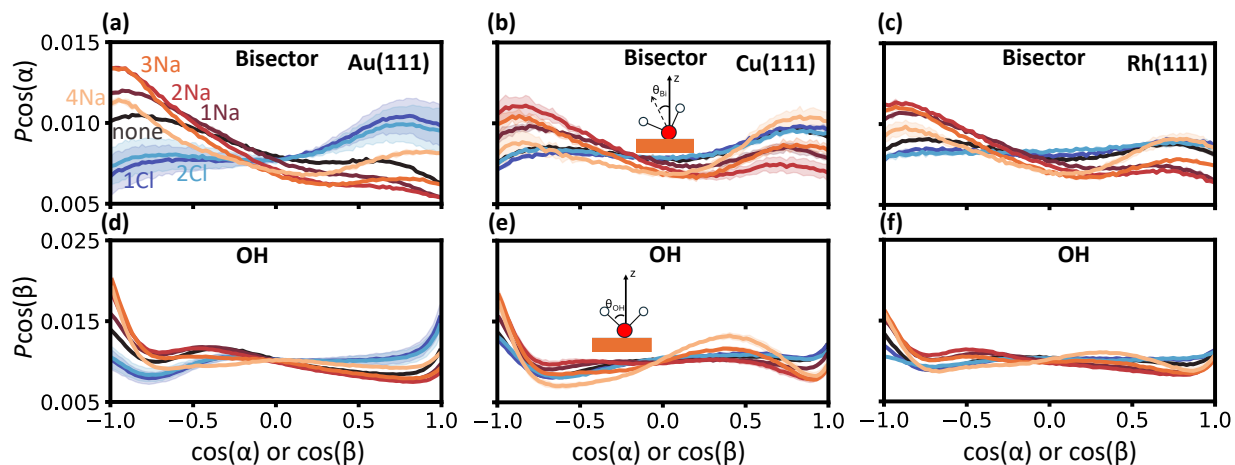


Figure S21: Orientational distribution of the second water layer measured as the probability distribution of the cosines of the angle between **(a-c)** the bisector of the water and the surface normal and **(d-f)** between the O-H bonds of water and the surface normal for Au, Cu, and Rh (111) facets using MACE-MD. Each solid line is an average of three separate 2 ns runs, and the shaded region indicates one standard deviation away from the mean.

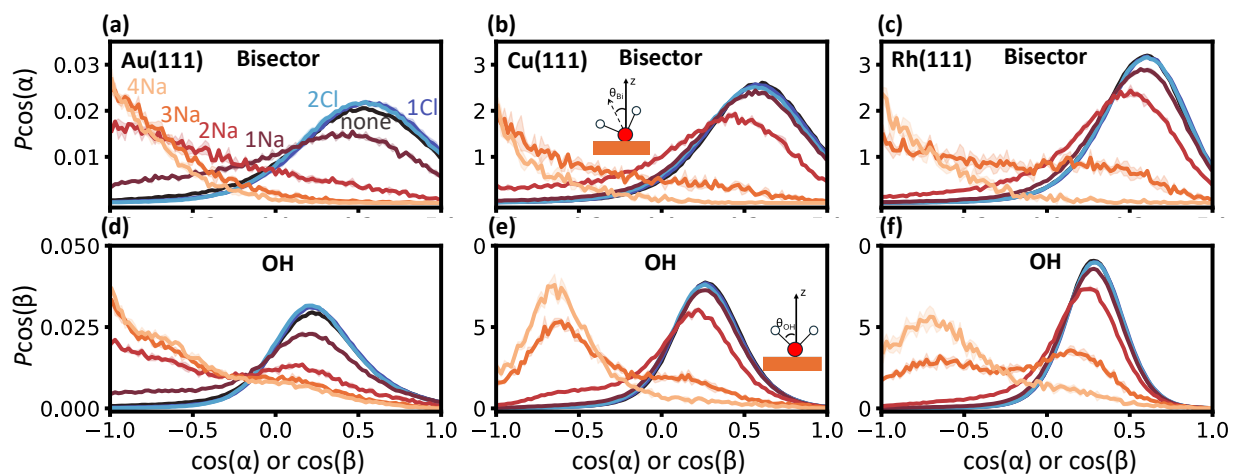


Figure S22: Orientational distribution of the chemisorbed water layer measured as the probability distribution of the cosines of the angle between **(a-c)** the bisector of the water and the surface normal and **(d-f)** between the O-H bonds of water and the surface normal for Au, Cu, and Rh (111) facets using MACE-MD. Each solid line is an average of three separate 2 ns runs, and the shaded region indicates one standard deviation away from the mean.

Vibrational density of states (VDOS) analysis results

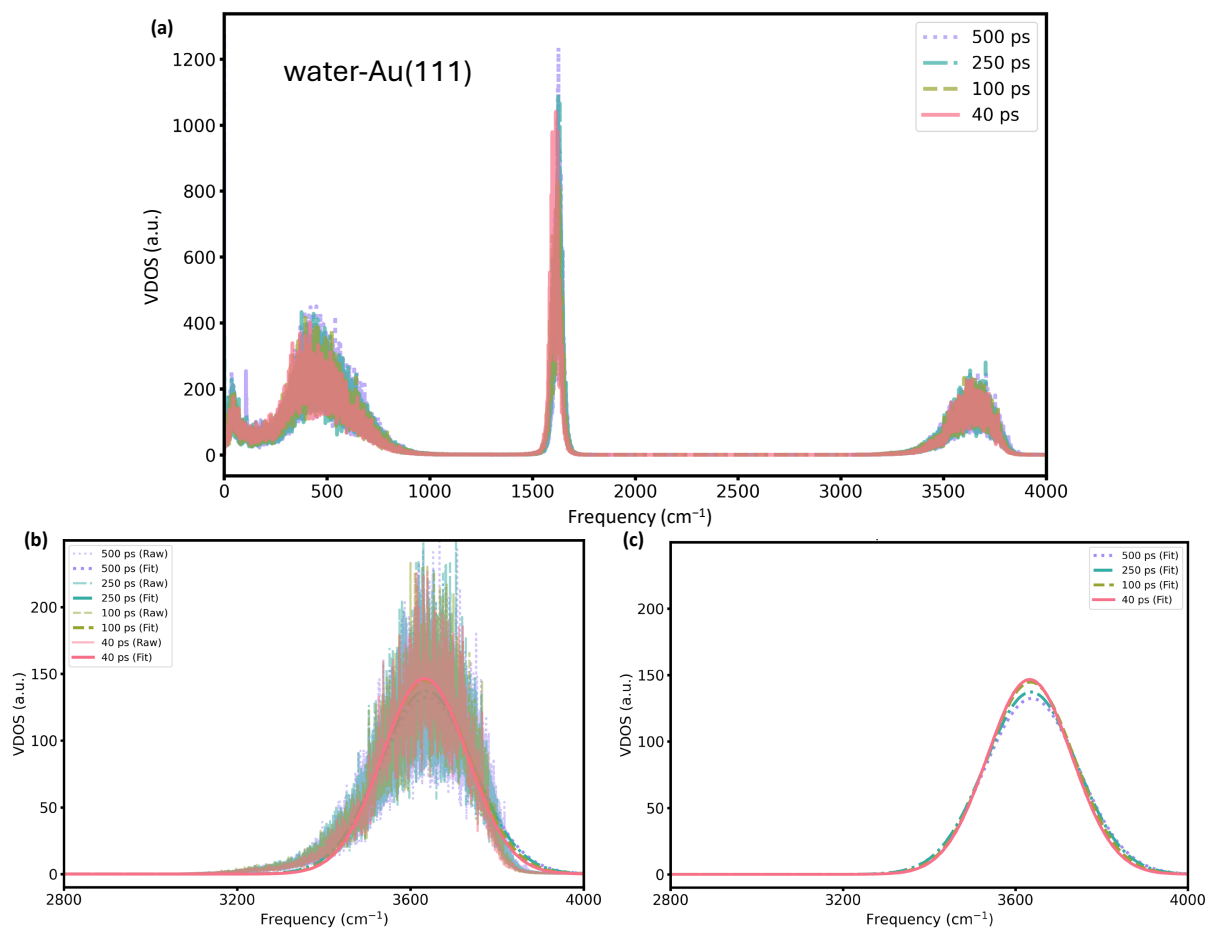


Figure S23: Vibrational density of states (VDOS) analysis of the near surface water with **(a)** different vibrational modes over frequency range of 0–4000 cm^{-1} and of the **(b-c)** O–H stretching at different simulation time for Au(111) pure water system ($\sigma = 0$).

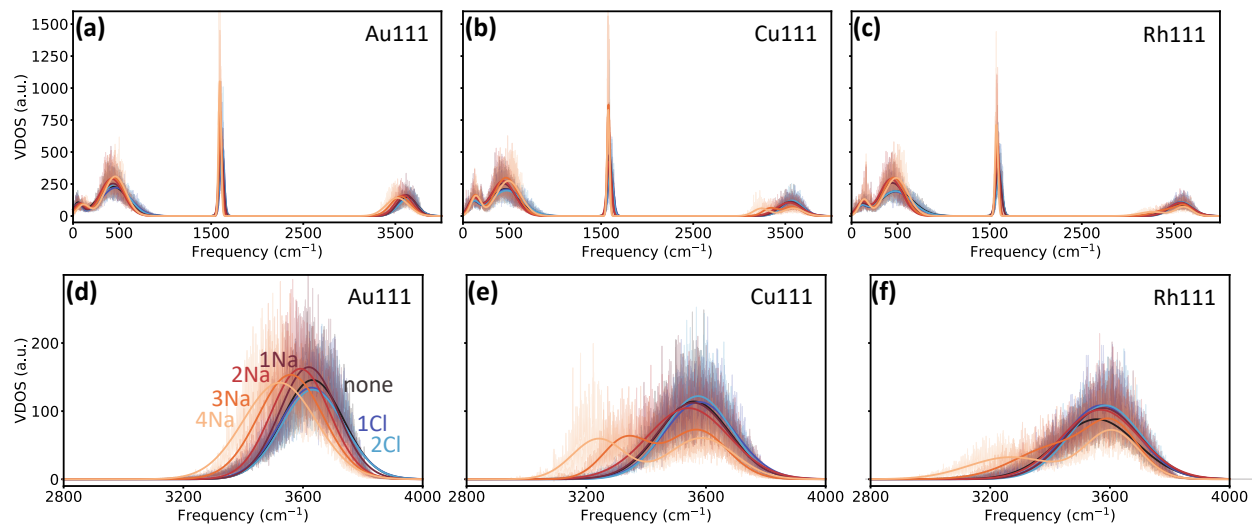


Figure S24: Vibrational density of states (VDOS) analysis of the near surface water with (a-c) different vibrational modes over frequency range of 0–4000 cm^{-1} and of the (d-f) O–H stretching. The VDOS analysis is conducted on a 40ps MACE MD simulation sampled every 1 fs and performed on final configuration from 2 ns simulation. Each solid line represent an average over three separate runs.

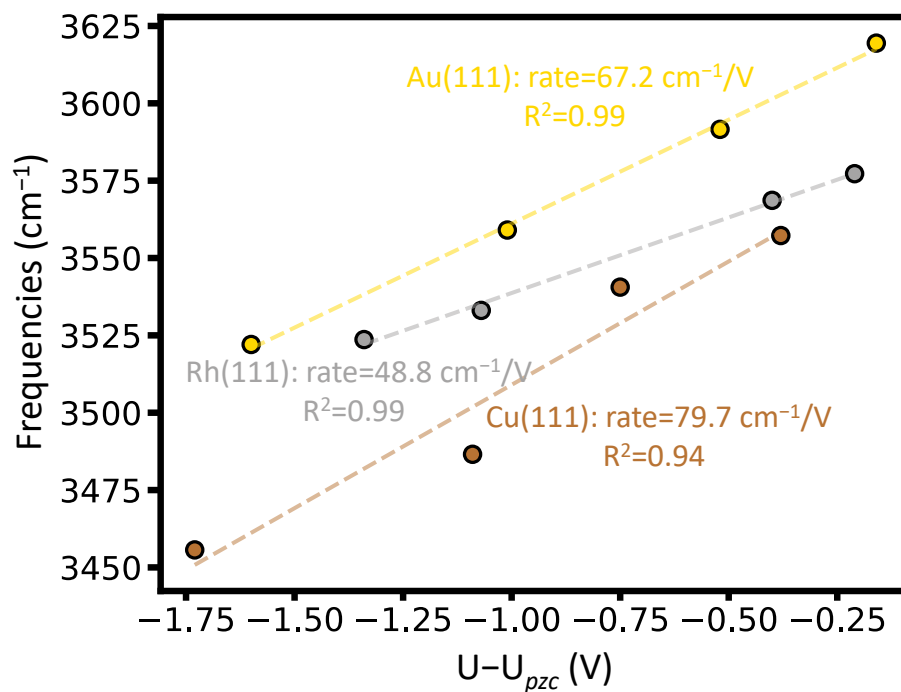


Figure S25: Mean O–H stretching frequency from vibration density of states against potential for systems with $U < U_{pzc}$. The linear fits provides Stark tuning rate.

Methods for MD trajectory analysis

Atomic density calculation

To characterize the interfacial structure of water and ions on the Au, Cu, and Rh electrodes, we computed atomic density profiles along the direction normal to the metal surface. The analysis was performed using a custom Python script,⁷ which incorporates the Atomic Simulation Environment (ASE),⁸ MDTraj,⁹ and SimSoliq¹⁰ libraries. SimSoliq was used to extract density data from each individual trajectory. For each system, density profiles from three independent simulations were averaged to obtain the ensemble mean.

Water orientation calculation

To quantitatively characterize the orientational ordering of water molecules within the metal–electrolyte interfacial region, we analyzed the angles formed by the water molecules relative to the axis normal (z -direction) to the metal surface. The analysis was conducted frame by frame for three independent trajectories to ensure robust statistical sampling. The methodology is broken down into three main steps: (1) identification of interfacial water molecules, (2) calculation of the orientational parameters, and (3) data aggregation. First, water molecules residing in a specific layer of interest (e.g., the first solvation layer) were identified in each trajectory frame. This selection was based on the position of the oxygen atom along the z -direction. A spatial region was defined by a lower and upper z -coordinate boundary, and only water molecules with oxygen atoms inside this region were included in the analysis. Second, for each water molecule in this region, we calculated the cosine of the angle between the surface normal and two key molecular vectors: (i) the molecule’s bisector vector ($\cos(\alpha)$), to determine the overall pointing direction, and (ii) each of its O-H bond vectors ($\cos(\beta)$), to assess hydrogen bonding directionality. Finally, these cosine values were collected across all frames and all replicate trajectories to generate probability distributions, which reveal the dominant orientational preferences of water at the interface. For benchmarking, we compared only the angular distribution of water molecules in the first layer. During the production MD runs, orientations were recorded for the chemisorbed, first-layer, and second-layer waters. The oxygen–surface distance criteria used to define each water layer are provided in Table S4.

Determining the coordination number of ions

Ion coordination numbers (CN_{ion}) were determined by counting the number of water oxygen atoms (O_w) within the first coordination shell, defined by the first peak in the ion– O_w radial distribution function ($g(r)$). For each ion species, CN_{ion} values were obtained by averaging over 2 ns of MACE–MD trajectories from three independent simulations. The first-shell cutoff distances were 3.4 Å for Na^+ and 5.0 Å for Cl^- . Bulk coordination numbers were computed using MD frames from trajectory in which the ion was positioned at least 5 Å away from the metal surface, whereas interfacial coordination numbers were computed from frames where the ion was within 5 Å of the surface.

Sample pictures of all systems with different surface charge density

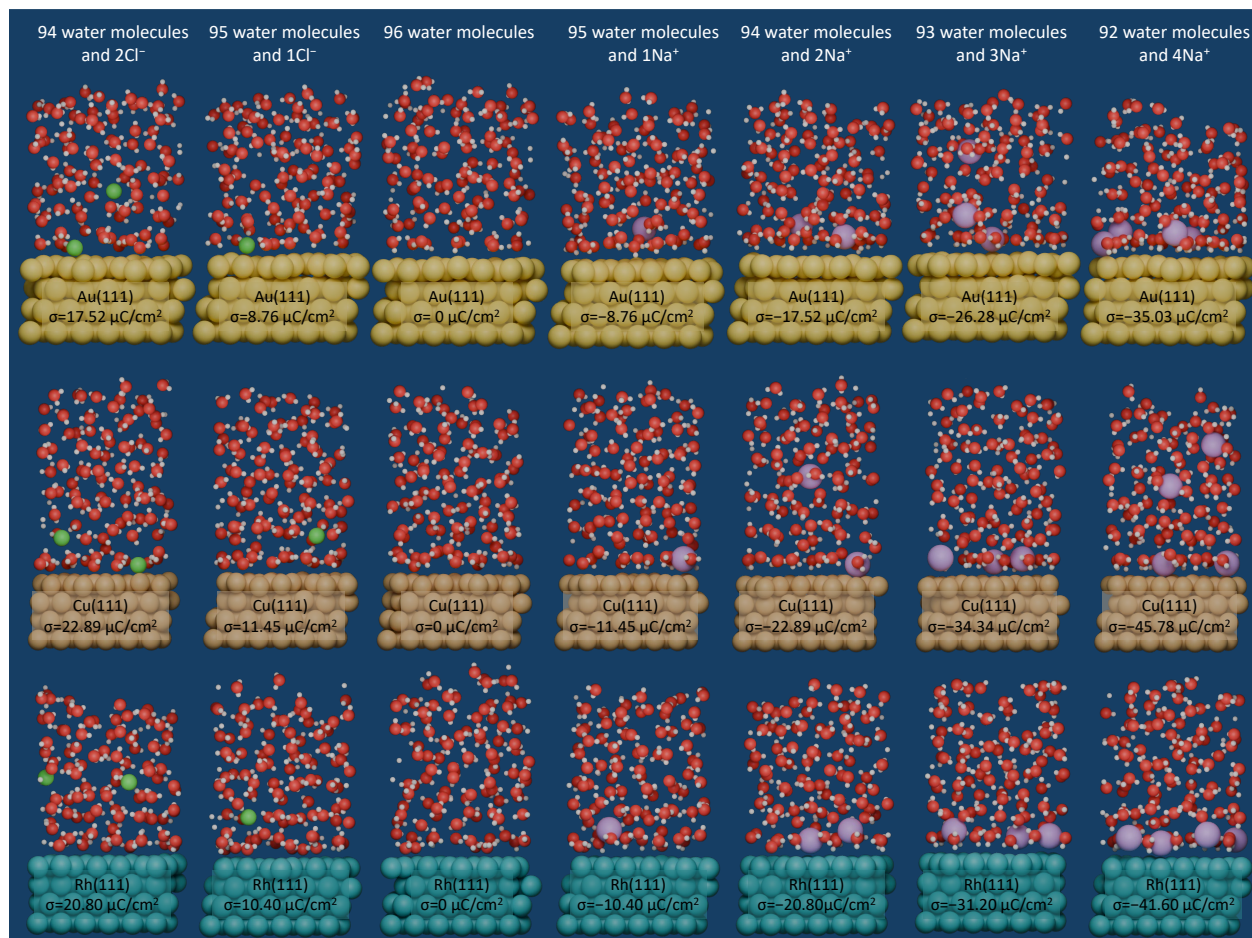


Figure S26: Sample pictures of different systems in decreasing order of their surface charge density.

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